ADVANCED FUNCTIONAL MATERIALS

FULL PAPER

High Mechanical Strength Double-Network Hydrogel with Bacterial Cellulose**

By *Atsushi Nakayama*, *Akira Kakugo*, *Jian Ping Gong*,* *Yoshihito Osada*, *Mitsuo Takai*, *Tomoki Erata*, and *Shin Kawano*

Double-network (DN) hydrogels with high mechanical strength have been synthesized using the natural polymers bacterial cellulose (BC) and gelatin. As-prepared BC contains 90 % water that can easily be squeezed out, with no more recovery in its swelling property. Gelatin gel is brittle and is easily broken into fragments under a modest compression. In contrast, the fracture strength and elastic modulus of a BC–gelatin DN gel under compressive stress are on the order of megapascals, which are several orders of magnitude higher than those of gelatin gel, and almost equivalent to those of articular cartilage. A similar enhancement in the mechanical strength was also observed for the combination of BC with polysaccharides, such as sodium alginate, gellan gum, and ι-carrageenan.

1. Introduction

Hydrogels are composed of a three-dimensional hydrophilic polymer network in which a large amount of water is interposed. Due to their unique properties, a wide range of medical, pharmaceutical, and prosthetic applications have been proposed.[1–5] However, most hydrogels suffer from a lack of mechanical toughness, and only a few applications have been realized. In our previous paper, we reported on a novel method for obtaining hydrogels with an extremely high mechanical toughness.[1] The hydrogels were composed of two kinds of hydrophilic polymer, forming a double-network (DN) structure, which, despite containing 90 % water, had an elastic modulus of 0.4–0.9 MPa and exhibited a compressive fracture strength as high as a few to several tens of megapascals.

The point of this discovery is that DN is composed of a combination of a stiff, brittle first network, and a soft, ductile second network. The molar composition and the cross-linking ratio of the first and second networks are also important. This DN gel is completely different in concept from traditional interpenetrating polymer networks (IPN)[6–9] or fiber-reinforced hydrogels, which are only linear combinations of two component networks.

Herein we describe, on the basis of this principle, the synthesis of biocompatible hydrogels with a high mechanical strength by using the natural polymers gelatin and bacterial cellulose (BC). Bacterial cellulose is an extracellular cellulose, produced by bacteria of genus *Acetobacter* and consisting of a hydrophobic, ultrafine-fiber network stacked in a stratified structure.[10] Due to this unique structure, it shows mechanical anisotropy, with a high tensile modulus (2.9 MPa) along the fiber-layer direction but a low compressive modulus (0.007 MPa) perpendicular to the stratified direction. In an as-prepared BC containing 90 % water, however, the water is easily squeezed out, and there is no more recovery in the swelling property due to hydrogen-bond formation between cellulose fibers. Gelatin is a polypeptide derived from an extracellular matrix, collagen. Gelatin gel can retain water and recovers from repeated compression. Due to its poor mechanical strength, a gelatin gel is broken quite easily into many fragments simply by pressing with a finger. Our strategy is to create a DN gel consisting of BC and gelatin that has a reversible swelling ability after repeated compression.

2. Results and Discussion

To obtain a BC–gelatin DN gel, BC was immersed in gelatin solution, then the gelatin was cross-linked by *N*-(3-dimethylaminopropyl)-*N*′-ethylcarbodiimide hydrochloride (EDC). EDC is one of the zero-length cross-linkers that modify amino acid side groups, in order to permit cross-linking without any carbon atoms between reaction moieties. As described in the previous section, when pressed in the direction perpendicular to the stratified structure of the BC gel, water is easily squeezed out from the gel, which does not recover to its initial swollen state (Fig. 1, first column). The chemically cross-linked gelatin gel is isotropic and mechanically brittle. It is easily broken into fragments under a modest compression of 0.1 MPa (Fig. 1, second column). In contrast, the BC–gelatin DN gel can hold water even under pressures as high as 3.7 MPa and completely recovers its original shape even after repeated compression in the direction perpendicular to the stratified struc-

[*] Prof. J. P. Gong, A. Nakayama, Dr. A. Kakugo, Prof. Y. Osada
Graduate School of Science, Hokkaido University
Sapporo 060-0810 (Japan)
E-mail: gong@sci.hokudai.ac.jp
Prof. J. P. Gong
Presto, JST
Sapporo 060-0810 (Japan)
Prof. M. Takai, Dr. T. Erata, Dr. S. Kawano
Graduate School of Engineering, Hokkaido University
Sapporo 060-0810 (Japan)

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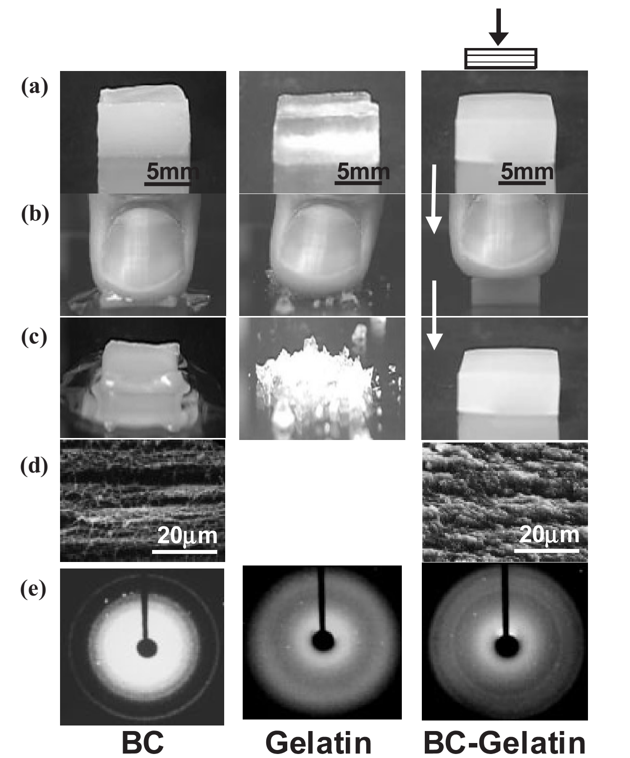

Figure 1. Pictures of the BC, gelatin, and BC–gelatin DN gels: a) before compression, b) during compression, and c) after 10 min compression. d) Scanning electron microscopy images of the stratified structure of the BC and BC–gelatin DN gels. e) Wide-angle X-ray diffraction patterns of the BC, gelatin, and BC–gelatin DN gels.

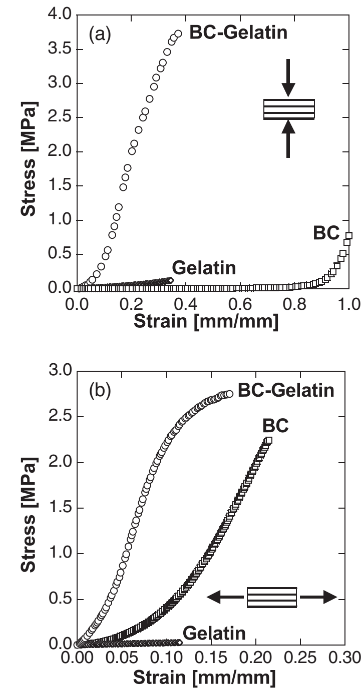

Figure 2. a) Compressive and b) elongational stress–strain curves of BC–gelatin DN, gelatin, and BC gels. The compression and elongation were performed perpendicular to and parallel to the stratified direction of the BC and BC–gelatin DN gels, respectively. The concentration of gelatin in the feed was 30 wt.-%, while the EDC concentration was 1 M.

ture of the BC–gelatin gel (Fig. 1, third column). Figure 2a shows typical compressive stress–strain curves of BC, gelatin, and their composite BC–gelatin DN gels. The compression was applied in the direction perpendicular to the stratified structure of the BC and BC–gelatin gels. The gelatin gel and BC–gelatin DN gel were prepared from a 30 % gelatin solution in the presence of 1 M EDC. One can see that the BC–gelatin DN gel shows a different stress–strain profile compared with individual BC and gelatin gels. The composite gel shows a compressive elastic modulus of 1.7 MPa in the direction perpendicular to the stratified structure, which is more than 240 times higher than that of BC (0.007 MPa), and almost eleven times than that of the gelatin gel (0.16 MPa) (Table 1). The fracture strength of the BC–gelatin DN gel in compression in the direction perpendicular to the stratified structure reached 3.7 MPa. This value is about 31 times higher than that of gelatin gel (0.12 Mpa; Table 1).

Figure 2b shows the tensile stress–strain curve of gels along the direction of the stratified layer. BC–gelatin DN gel can sustain nearly 3 MPa, and has an elastic modulus of 23 MPa, which is 112 times larger than that of gelatin gel.

A cyclic compressive test demonstrated that BC did not retain its original stress–strain profile, whereas the BC–gelatin DN gel recovered well from a second compressive deformation up to 30 % strain. Thus, the DN structure led to a substantial improvement of the mechanical properties.

Table 1. Effect of gelatin concentration in the feed on the degree of swelling (q) and the elastic modulus (E), fracture stress (σ), and fracture strain (λ) of BC–gelatin, gelatin, and BC gels in compression and elongation (–, not measured). The EDC concentration was 1 M.

		Compression			Elongation		
Sample	q	E [MPa]	σ [MPa]	λ [mm/mm]	E [MPa]	σ [MPa]	λ [mm/mm]
BC–gelatin							
15 wt.-%	11	1.3	2.1	0.41	5.0	1.1	0.22
30 wt.-%	5.8	1.7	3.7	0.37	23	2.8	0.17
40 wt.-%	3.6	2.8	5.2	0.36	19	3.7	0.23
50 wt.-%	3.1	3.9	5.3	0.44	21	3.8	0.28
gelatin							
15 wt.-%	41	0.047	0.029	0.34	0.040	0.006	0.11
30 wt.-%	11	0.16	0.12	0.35	0.23	0.025	0.11
40 wt.-%	5.2	0.93	0.44	0.29	0.78	0.070	0.10
50 wt.-%	3.8	1.2	1.2	0.37	1.8	0.18	0.10
BC	120	0.007	–	–	2.9	2.2	0.21

Similar improvement in the mechanical strength was also observed for combinations with polysaccharides, such as sodium alginate, gellan gum, and ι-carrageenan (Table 2). For example, for the BC–ι-carrageenan DN gel, the compressive elastic modulus was 0.12 MPa, which is 17 and 13 times higher than that of individual BC and ι-carrageenan gels, respectively.

As shown in the scanning electron microscopy (SEM) image of Figure 1d, BC consists of alternating dense and sparse layers consisting of cellulose fibers with a period of about 10 μm, which contributes to the anisotropic mechanical strength. One

Table 2. Degree of swelling (q) and the elastic modulus (E), fracture stress (σ), and fracture strain (λ) of BC–polysaccharides: BC–sodium alginate, BC–gellan gum, and BC–ι-carrageenan DN gels in compression and elongation (–, not measured).

Sample	q	Compression			Elongation		
		E [MPa]	σ [MPa]	λ [mm/mm]	E [MPa]	σ [MPa]	λ [mm/mm]
BC–gellan gum	27	0.38	–	–	2.3	1.2	0.37
BC–sodium alginate	20	0.61	–	–	6.7	2.2	0.30
BC–ι-carrageenan	30	0.12	–	–	1.8	0.5	0.26
gellan gum	42	0.62	0.47	0.60	1.4	0.16	0.11
sodium alginate	22	0.14	–	–	0.37	0.6	0.89
ι-carrageenan	158	0.009	–	–	–	–	–

can see that in the BC–gelatin DN gel, the gelatin fills in the cellulose-poor layer and shows a more homogeneous texture. Wide-angle X-ray diffraction (WAXD) shows the crystalline pattern of BC, which is well-maintained in the BC–gelatin DN gel (Fig. 1e). Gelatin gel is structurally amorphous and shows no X-ray diffraction pattern.

For a material that is composed of a layered structure with two alternative layers of elastic moduli E_s and E_h ($E_s \ll E_h$), the macroscopic modulus E of the material perpendicular to the stratified direction depends only on the modulus of the soft component, E_s, and not on the modulus of the hard component, E_h, as given by[11]

$$E = \frac{d_s + d_h}{d_s} E_s \tag{1}$$

where E is the macroscopic modulus of the material perpendicular to the stratified direction; E_s and E_h are the moduli of the soft and hard components, respectively; and d_s and d_h are the thicknesses of the soft and hard components, respectively. Therefore, if we consider that cellulose-rich parts behave as the hard layer and gelatin-rich parts behave as the soft layer, the overall macroscopic modulus should largely depend on the modulus of the soft component, gelatin, since $d_s \approx d_h$. The mechanical property of gelatin can be modulated by the extent of crosslinking with EDC.

Therefore, we investigated the effect the degree of crosslinking had on the elastic modulus by changing the EDC concentration. Without EDC, a 30 wt.-% gelatin aqueous solution cannot form a gel that is strong enough for mechanical measurement. By chemically crosslinking with EDC, gelatin formed a solid gel with a compressive elastic modulus of about 0.1 MPa, as shown in Figure 3. As shown in Figure 3a, a substantial increase in the compressive elastic modulus of BC-gelatin DN gel was observed with increasing concentration of cross-linker, while only a modest enhancement of the elastic modulus was observed in gelatin gel. It was also found that the compressive fracture stress of BC-gelatin DN gel increases with increasing EDC concentration, similar to the behavior observed for gelatin gel (Fig. 3b). Thus, the mechanical properties of the DN gel is predominantly enhanced by the crosslinkage of gelatin.

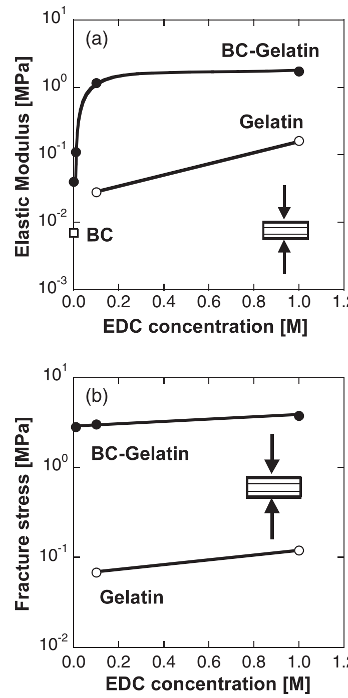

Figure 3. Effect of the EDC crosslinker concentration on a) the compressive elastic modulus, and b) the compressive fracture stress of BC–gelatin DN, gelatin, and BC gels. The compression was performed in the direction perpendicular to the stratified structure of the BC and BC–gelatin gels. The concentration of gelatin in the feed was 30 wt.-%.

Since the mechanical properties of a hydrogel should also depend on its water content, that is, the degree of swelling, q, we investigated the effect of q on the elastic modulus and fracture stress of the BC–gelatin gel. As shown in Figures 4a,b, both the elastic modulus and fracture stress decrease with increasing degrees of swelling, following a power-law relation. The fracture stress of the BC–gelatin gel was 4–18 times higher than that of the gelatin gel over the same range of degree of swelling. Also, the elastic modulus of the BC–gelatin gel was 2–8 times higher than that of the gelatin gel. Both the elastic modulus and fracture stress of the BC–gelatin gel were less sensitive to the degree of swelling than was the gelatin gel. Thus, despite its high water content (70 wt.-%), the BC–gelatin gel exhibited high mechanical strength: a fracture stress of 5 MPa and an elastic modulus of 4 MPa, both of which are almost as high as those of articular cartilage.[12]

In our previous paper, we emphasized that, in order to obtain gels with a high mechanical strength, two structural parameters are crucial.[1] One is the unit ratio of the second network to the first network (hereafter denoted as R), and the other is the cross-linking density of the networks. Figure 5 shows the effect of the unit ratio of the amino acid of the second network (gelatin) to the glucose of the first network (BC) on the elastic modulus and fracture stress. R was determined by elemental analysis of the nitrogen and carbon of gels prepared using various gelatin concentrations (Table 3). A dramatic enhancement in the mechanical strength of the gel was obtained when R exceeded eleven for the BC–gelatin gel. Thus, the mechanical

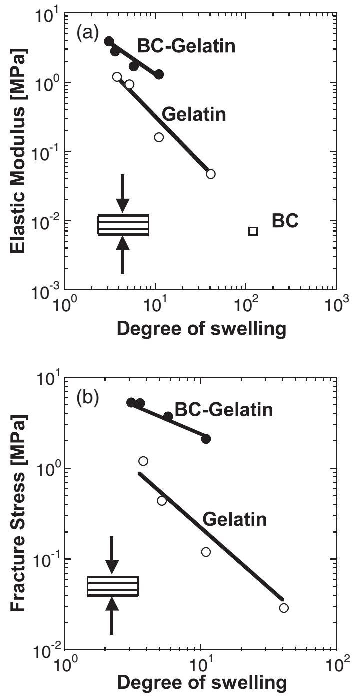

Figure 4. Dependence of a) the compressive elastic modulus; and b) the compressive fracture stress on the degree of swelling q for BC–gelatin DN, gelatin, and BC gels. The compression was performed in the direction perpendicular to the stratified structure of the BC and BC–gelatin gels.

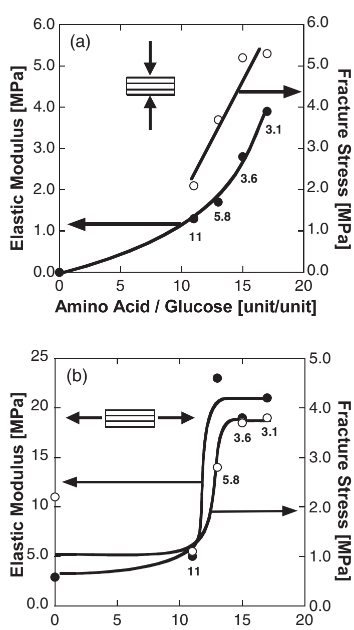

Figure 5. Dependence of the elastic modulus and fracture stress on the ratio R of gelatin amino acid to BC glucose in a) compression and b) elongation. Numbers in the figure represent the degree of swelling, q, of the gels.

properties are dominantly determined by the amount of gelatin, which is the soft component.

Table 3. Unit ratio R of gelatin amino acid to BC glucose for BC–gelatin DN gels prepared at various gelatin concentrations; the EDC concentration was 1 M.

Concentration [wt.-%]	Amino acid/glucose [unit/unit]
15	11
30	13
40	15
50	17

The friction coefficient of the BC–gelatin gel was measured against a glass substrate at a sliding velocity of 100 mm min^{-1} by a tribometer. It was found that under a normal pressure of 1 MPa, the gel has a friction coefficient of 10^{-3}, which is as low as that of articular cartilage.

3. Conclusions

We succeeded in creating a hydrogel from natural polymers, which not only has a mechanical strength as high as several megapascals, but also has a low frictional coefficient of the order of 10^{-3}. This hydrogel might be a good candidate as an artificial articular cartilage. These gels might open up a new era of soft and wet materials as substitutes for articular cartilage and other tissues.

4. Experimental

Materials: Gelatin powder isolated from calf skin was purchased from Junsei Chemical Co., Ltd., and used without further purification. *N*-(3-dimethylaminopropyl)-*N′*-ethylcarbodiimide hydrochloride (EDC). and 3-(*N*-morpholino)propanesulfonic acid (MOPS; used to adjust the pH) were purchased from Tokyo Kasei Co., Ltd. and Sigma Chemical Inc., respectively.

Gel Preparation: *Acetobacter xylinum* (American Type Culture Collection 53582) was cultured in Hestrin–Schramm medium at pH 6.0, and incubated at 28 °C for 10 days. A disc-shaped bacterial cellulose (BC) obtained after cultivation, about 10 mm thick and 50 mm in diameter, was washed with a 1 % aqueous NaOH solution to remove the alkali-soluble components; it was then washed with water for a prolonged time at room temperature. (A detailed description is given elsewhere [13].) The purified BC gel was immersed in an aqueous solution of gelatin (pH 7.0) at 50 °C for one week and then immersed in an aqueous EDC solution for 4 days at room temperature to allow chemical crosslinking to occur. The concentration of gelatin was varied from 15–50 wt.-%, and its pH was adjusted using MOPS. The EDC concentration was varied from 0–1.0 M. After crosslinking of gelatin with EDC, the samples were washed with a large amount of deionized water for one week at room temperature to equilibrate and wash away the residual chemicals.

The gelatin gel was prepared as follows: gelatin was dissolved in a MOPS buffer, at a concentration in the range of 15–50 wt.-%, and left for a week at 50 °C. Then the gelatin solution was kept at room temperature to form a physically cross linked gelatin gel. Finally, it was immersed in an aqueous solution of EDC (1 M) for 4 days at room temperature.

The amount of water contained in the gels was characterized by the degree of swelling, q, which is defined as the weight ratio of swollen to dry sample. Dry gels were obtained by drying in vacuo until a constant weight was reached.

The BC–polysaccharide double-network (DN) gel was prepared as follows: the purified BC gel was immersed in an aqueous solution of polysaccharide (sodium alginate, gellan gum, or ι-carrageenan) at 70 °C for a week, and then the gel was immersed in an aqueous $CaCl_2$ solution for 4 days at room temperature as a physical cross-linking reaction. Samples were washed with a large amount of deionized water for a week at room temperature to equilibrate and wash away the residual chemicals.

The unit ratio (R) of the amino acid derived from gelatin to the glucose derived from BC was determined by the elemental analysis of nitrogen and carbon in the gels. The amount of gelatin in the BC–gelatin DN gel was estimated from the total nitrogen content of gelatin as 18 % [14]. The molecular weights of the amino acid and glucose were 100 (average of the constituent amino acids in gelatin) and 180, respectively.

Compression and Tensile Testing: A tensile-compressive tester (TENSILON, Orientec Co.) was used to measure the mechanical properties of the gels. For the compression tests, samples were cut into discs (10 mm in diameter and 5 mm thick) and compressed perpendicular to the BC layers by two parallel metal platens connected to a load cell, at a strain rate of 10 % min^{-1} at room temperature. For the tensile tests, samples were cut into dumbbell shapes (dowel width, thickness, and length of 2, 2, and 10 mm, respectively, and bell width, thickness, and length of 10, 2, and 7 mm, respectively) and stretched parallel to the BC layers using a clamp attachment at a strain rate of 10 % min^{-1}. Strain rates were referenced to the initial thickness or length of the specimens. Failure points of compressive and tensile tests were determined from the peak of the stress–strain curve. The elastic modulus, E, was determined by the average slope over the strain ratio range of 0–0.1 of the strain ratio from the stress–strain curve.

Friction Measurements: Friction measurements were performed using gel samples equilibrated in distilled water against a glass substrate at room temperature using a tribometer (Heidon 14S/14DR, Shindom Sci. Co.). The gels were 10 mm by 10 mm and 5 mm thick. The sample was embedded in a square frame of adjustable size and pressed against a glass surface that was fixed on the lower board. The glass surface was wetted with distilled water and was driven to move horizontally and repeatedly in a velocity range of 100 mm min^{-1} over a distance of 100 mm. Prior to the measurement, the glass plate was carefully washed, rinsed in distilled water, and dried in air. The frictional coefficient, μ, is calculated by $\mu = F/W$, where F is the frictional force and W is the normal load. The detailed description of measurement is given elsewhere [15].

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